



ST BENEDICT'S

SUBJECT	Chemistry leBT Exam
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mr McCoy
DATE	23 July 2025
MARKS	11
DURATION	30 minutes

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 10 multiple-choice questions on 6 pages (including this front page). A Data Sheet of 2 pages (i–iii) is provided. Please check that your question paper is complete.
2. Fill in the circles completely with HB pencil
3. Please colour on the appropriate circle: ☐ A ☐ B ☒ C ☐ D
Please DO NOT: ☐ ☒ ☒ ☒
Put a line or cross or partially colour in.
To make a change, erase completely and fill in the correct circle.
4. Read all the questions carefully.
5. Answer all the questions.
6. **No calculators may be used in this paper.**
7. All questions are equally weighted with no negative marking.
8. Learners answering this examination on paper will require a B pencil and an eraser.

	Total
Mark	
Question Total	

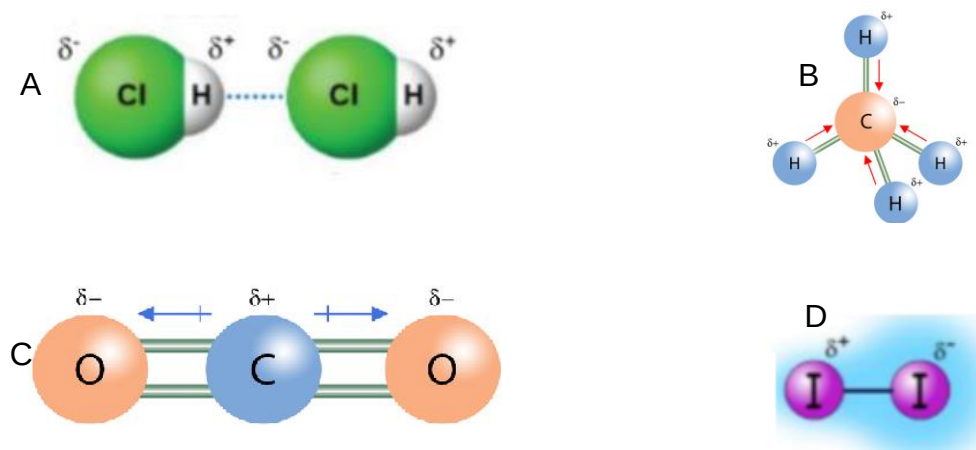
Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1

Consider the following substances:

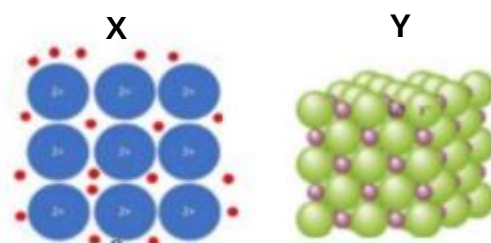


Which of the following pair is **incorrect**?

	Intramolecular	Intermolecular force
A	Polar covalent	Dipole-dipole forces
B	Polar covalent	London forces
C	Polar covalent	Dipole-dipole forces
D	Non-polar covalent	London forces

QUESTION 2

Which of the following pair is **correct**?



	Substance	Intramolecular	Force
A	X	Ionic	Electrostatic attraction between cations and anions
B	Y	Ionic	Electrostatic force of attraction, positive kernels and delocalised electrons
C	X	Metallic	Electrostatic force of attraction, positive kernels and delocalised electrons
D	Y	Metallic	Electrostatic attraction between cations and anions

QUESTION 3

Consider the boiling points of the two compounds in the table below.

Substance	Boiling point (°C)
HF	19,5
NH ₃	-33

Which option best describes the difference in boiling points?

- A The boiling point of HF is greater than that of NH₃ as the hydrogen bonding in HF is more polar than in NH₃.
- B The boiling point of HF is greater than that of NH₃ as the HF forms less hydrogen bonds than NH₃.
- C The boiling point of HF is greater than that of NH₃ as the dipole-dipole forces in HF is stronger than the hydrogen bonding in NH₃.
- D The boiling point of HF is greater than that of NH₃ as the HF forms more hydrogen bonds than NH₃.

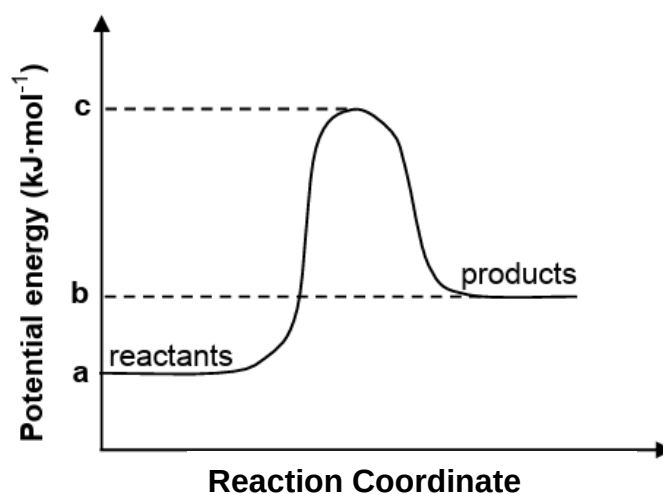
QUESTION 4

An unknown metal **M** forms the compound **M(SO₄)₂**. In which of the following is the valency for **M** is correct?

- A M(NO₃)₂
- B M₂O
- C M₃PO₄
- D M₃N₄

QUESTION 5

The potential energy graph for a hypothetical chemical reaction is shown below.



What type of reaction is taking place and what are the correct methods to calculate ΔH and E_a ?

	Type of reaction	ΔH	E_a
A	Endothermic	$b - a$	$c - b$
B	Endothermic	$b - a$	$c - a$
C	Endothermic	$a - b$	$a - c$
D	Exothermic	$a - b$	$b - c$

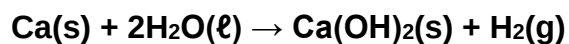
QUESTION 6

How many hydrogen atoms are present in 8,5 g of NH_3 ?

- A $3,01 \times 10^{23}$
- B $6,02 \times 10^{23}$
- C $9,03 \times 10^{23}$
- D $2,40 \times 10^{24}$

QUESTION 7

A 40 g piece of impure calcium metal is placed in excess water in a closed system. The balanced chemical equation for the reaction is shown below:

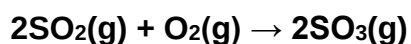


At the end of the reaction 0,75 moles of H_2 was collected. The excess water was then evaporated and collected and this equalled 0,25 moles. Select the appropriate combination for the percentage purity of the calcium metal and the amount of water initially added to the impure calcium metal.

	% PURITY OF CALCIUM METAL	AMOUNT OF WATER ADDED INITIALLY (mol)
A	75	1,75
B	50	1,50
C	25	1,75
D	75	1,50

QUESTION 8

In the Contact process, the following reaction takes place:



Initially, 2 000 dm³ of SO₂ and 1 000 dm³ of O₂ are placed in the container. At the end of the reaction the yield of SO₃ gas is 50%. Pressure and temperature remain constant throughout the process. The total volume of gas present inside the container now will be:

- A 1 500 dm³
- B 2 500 dm³
- C 4 000 dm³
- D 5 000 dm³

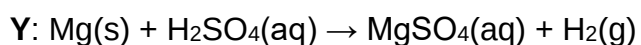
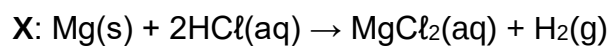
QUESTION 9

Which ONE of the following statements is TRUE for an oxidising agent?

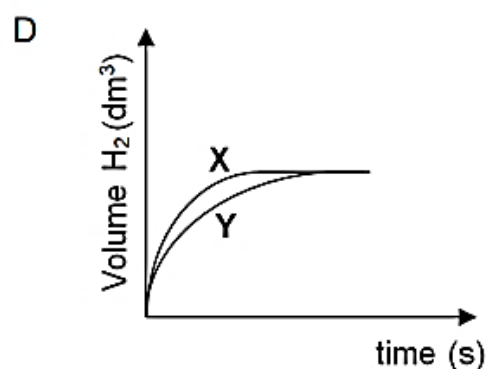
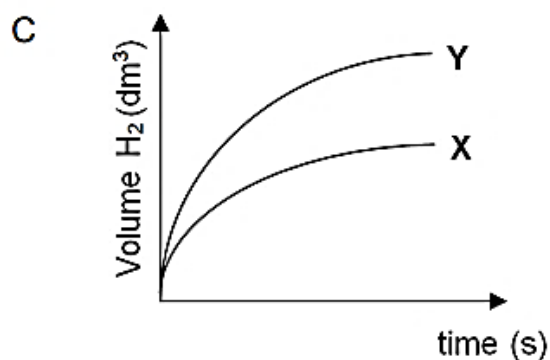
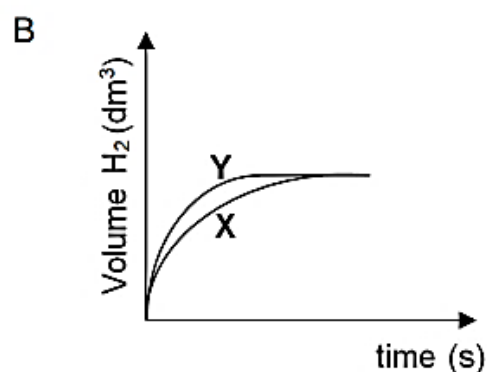
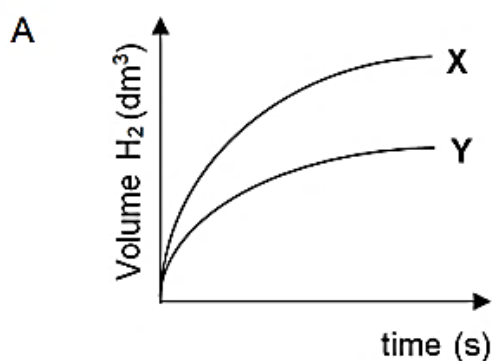
- A It gains electrons.
- B It causes another species in the reaction to be reduced.
- C Its oxidation number does not change during a chemical reaction.
- D Its oxidation number increases during a chemical reaction.

QUESTION 10

Two pieces of magnesium ribbon of the exact same mass are reacted with 250 ml dilute solutions of hydrochloric acid and sulfuric acid respectively. The concentration of both solutions is $0,1 \text{ mol} \cdot \text{dm}^{-3}$. There is a piece of magnesium ribbon left after completion of the respective reactions. The balanced equations for the two reactions are shown below.



Which one of the following graphs (not drawn to scale) best describes the relationship between the amount of hydrogen gas produced over time for each reaction?



END OF ASSESSMENT